

Brutal Series - Paper B03 (Class 7): Nutrition in Plants, Integers, Physical & Chemical Changes, Perimeter & Area

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A leaf makes 6 mg of glucose each hour in bright sun. It gets 7 hours of sun, and the plant uses one-third of that glucose for its own respiration. How much glucose is stored, and which gas did the leaf take in to make it? [\[Nutrition in Plants\]](#)
- (A) 21 mg; carbon dioxide
 - (B) 35 mg; carbon dioxide
 - (C) 28 mg; oxygen
 - (D) 14 mg; carbon dioxide
 - (E) 28 g; carbon dioxide
 - (F) 42 mg; carbon dioxide
 - (G) 28 mg; carbon dioxide
2. At dawn the temperature was -6 deg C. It rose 3 deg C every hour for 4 hours, then fell 5 deg C. What is the final temperature? [\[Integers\]](#)
- (A) -1 deg C
 - (B) 11 deg C
 - (C) 2 deg C
 - (D) 6 deg C
 - (E) -11 deg C
 - (F) 1 deg C
3. A 50 g iron gate gains 8% of its mass as rust over a year. What is its new mass, and is rusting a physical or a chemical change? [\[Physical & Chemical Changes\]](#)
- (A) 4 g; chemical
 - (B) 50 g; chemical
 - (C) 46 g; chemical
 - (D) 54 g; physical
 - (E) 54 g; chemical
 - (F) 58 g; chemical
 - (G) 54 g; both
4. A rectangular garden is 18 m long and 12 m wide. A path runs all around the outside, adding 2 m to every edge (so 4 m extra on the length and 4 m on the width). What is the area of the larger rectangle including the path? [\[Perimeter & Area\]](#)
- (A) 352 sq m
 - (B) 136 sq m
 - (C) 330 sq m
 - (D) 216 sq m
 - (E) 440 sq m
 - (F) 256 sq m
 - (G) 308 sq m

5. A leaf has 300 stomata in every square millimetre. A patch of leaf measures 4 square millimetres. How many stomata is that, and on which surface of the leaf are most stomata usually found? [\[Nutrition in Plants\]](#)
- (A) 304; the lower surface
 - (B) 900; the lower surface
 - (C) 1200; both surfaces equally
 - (D) 120; the lower surface
 - (E) 1200; the upper surface
 - (F) 1200; the lower surface
 - (G) 75; the lower surface
6. A diver is at -18 m (below sea level). She rises 5 m, then dives down twice as far as she just rose. What is her final depth as an integer? [\[Integers\]](#)
- (A) -33 m
 - (B) -23 m
 - (C) -13 m
 - (D) -3 m
 - (E) -28 m
 - (F) -8 m
 - (G) 13 m
7. 200 g of ice melts completely into water. What is the mass of the water formed, and is melting a physical or a chemical change? [\[Physical & Chemical Changes\]](#)
- (A) 180 g; physical
 - (B) 0 g; physical
 - (C) 200 g; physical
 - (D) 200 g; chemical
 - (E) 220 g; physical
 - (F) 100 g; physical
8. A triangular park has a base of 40 m and a height of 25 m. Laying grass costs Rs 12 per square metre. What is the total cost to grass the park? [\[Perimeter & Area\]](#)
- (A) Rs 600
 - (B) Rs 6000
 - (C) Rs 3000
 - (D) Rs 1000
 - (E) Rs 6500
 - (F) Rs 500
 - (G) Rs 12000
9. A fungus growing on a dead leaf digests 12 g of a 30 g leaf by releasing juices onto it. What percent of the leaf is left undigested, and what is this mode of nutrition called? [\[Nutrition in Plants\]](#)
- (A) 40%; saprotrophic
 - (B) 60%; autotrophic
 - (C) 12%; saprotrophic
 - (D) 60%; insectivorous
 - (E) 18%; saprotrophic
 - (F) 60%; saprotrophic
 - (G) 60%; parasitic

10. Find the value of $(-4) \times (-3) \times (-2)$, then add 5 to the result. [\[Integers\]](#)

- (A) 24
- (B) -19
- (C) -24
- (D) 29
- (E) -29
- (F) 19
- (G) -14

11. A candle holds 40 g of wax. After burning for a while, 25 g of wax remains. How much wax burned away, and is burning a physical or a chemical change? [\[Physical & Chemical Changes\]](#)

- (A) 15 g; both
- (B) 65 g; chemical
- (C) 15 g; physical
- (D) 15 g; chemical
- (E) 10 g; chemical
- (F) 25 g; chemical
- (G) 40 g; chemical

12. A square field has a perimeter of 96 m. What is its area? [\[Perimeter & Area\]](#)

- (A) 384 sq m
- (B) 576 sq m
- (C) 9216 sq m
- (D) 144 sq m
- (E) 288 sq m
- (F) 2304 sq m
- (G) 48 sq m

13. A pitcher plant catches 5 insects on the first day, and the number it catches doubles each day. How many does it catch on the fourth day, and why does it trap insects at all? [\[Nutrition in Plants\]](#)

- (A) 15; to get nitrogen
- (B) 40; to get nitrogen
- (C) 20; to get nitrogen
- (D) 80; to get nitrogen
- (E) 40; to get sunlight
- (F) 40; to get carbon dioxide
- (G) 40; to get water

14. A submarine descends a total of 96 m in 8 equal downward steps. What integer describes one step, and is it positive or negative? [\[Integers\]](#)

- (A) -768 m, a negative number
- (B) 12 m, a positive number
- (C) -104 m, a negative number
- (D) -12 m, a negative number
- (E) -88 m, a negative number
- (F) -12 m, a positive number

15. Three spoons of sugar, each 5 g, dissolve in 200 g of water. What is the mass of the sugar solution, and is dissolving a physical or a chemical change? [\[Physical & Chemical Changes\]](#)

- (A) 200 g; physical
- (B) 215 g; chemical
- (C) 215 g; both
- (D) 15 g; physical
- (E) 185 g; physical
- (F) 205 g; physical
- (G) 215 g; physical

16. A room floor is 5 m by 4 m. Square tiles measure 50 cm by 50 cm. How many tiles are needed to cover the floor exactly? [\[Perimeter & Area\]](#)

- (A) 20 tiles
- (B) 40 tiles
- (C) 100 tiles
- (D) 800 tiles
- (E) 16 tiles
- (F) 80 tiles
- (G) 8 tiles

17. Bacteria in the root nodules of gram (a legume) fix nitrogen. If one hectare of gram adds 9 kg of nitrogen and a farmer plants 2.5 hectares, how much nitrogen is added, and what is the bacterium called?

[\[Nutrition in Plants\]](#)

- (A) 225 kg; Rhizobium
- (B) 18 kg; Rhizobium
- (C) 11.5 kg; Rhizobium
- (D) 2.25 kg; Rhizobium
- (E) 22.5 kg; Rhizopus
- (F) 22.5 kg; Lactobacillus
- (G) 22.5 kg; Rhizobium

18. On a number line a flea jumps from -7 to +5, then back to -2. What total distance does it cover, counting both jumps? [\[Integers\]](#)

- (A) 2
- (B) 26
- (C) 7
- (D) 19
- (E) 9
- (F) 5
- (G) 12

- 19.** Carbon dioxide turns clear lime water milky. If each test uses 6 mL of lime water and you run 7 tests, how much lime water is used, and turning milky shows which kind of change? [\[Physical & Chemical Changes\]](#)
- (A) 48 mL; chemical
 - (B) 42 L; chemical
 - (C) 13 mL; chemical
 - (D) 42 mL; physical
 - (E) 36 mL; chemical
 - (F) 42 mL; no change
 - (G) 42 mL; chemical
- 20.** A parallelogram has a base of 14 cm and a height of 9 cm. A second parallelogram has the same area but a base of 21 cm. What is the height of the second one? [\[Perimeter & Area\]](#)
- (A) 3 cm
 - (B) 4 cm
 - (C) 7 cm
 - (D) 6 cm
 - (E) 126 cm
 - (F) 9 cm
 - (G) 13.5 cm
- 21.** A dodder (Amarbel) twines on a host plant and sucks its food. If the host loses 15% of its 200 g of stored food to the dodder, how many grams are lost, and what kind of plant is the dodder? [\[Nutrition in Plants\]](#)
- (A) 170 g; a parasite
 - (B) 30 g; a host
 - (C) 15 g; a parasite
 - (D) 30 g; a saprotroph
 - (E) 30 g; a parasite
 - (F) 45 g; a parasite
 - (G) 20 g; a parasite
- 22.** Evaluate: $-20 + 6 \times (-3) - (-4)$. [\[Integers\]](#)
- (A) 34
 - (B) -42
 - (C) -30
 - (D) 42
 - (E) -38
 - (F) -34
- 23.** Iron sheets are coated with zinc to stop rust, a process called galvanisation. To coat 15 sheets, each needing 0.4 kg of zinc, how much zinc is needed, and what does the zinc prevent? [\[Physical & Chemical Changes\]](#)
- (A) 3 kg; rusting
 - (B) 0.6 kg; rusting
 - (C) 6 kg; burning
 - (D) 60 kg; rusting
 - (E) 15.4 kg; rusting
 - (F) 6 kg; melting
 - (G) 6 kg; rusting

- 24.** A circular track has a radius of 35 m. A runner jogs around it 3 times. How far does she run? (Take $\pi = \frac{22}{7}$.) [\[Perimeter & Area\]](#)
- (A) 330 m
 - (B) 220 m
 - (C) 770 m
 - (D) 440 m
 - (E) 110 m
 - (F) 660 m
 - (G) 630 m
- 25.** A plant releases 48 g of oxygen, and oxygen makes up 4 of every 5 parts of the gas it gives out (the rest is water vapour). How much gas does it release in total, and what process is this? [\[Nutrition in Plants\]](#)
- (A) 12 g; photosynthesis
 - (B) 60 g; transpiration
 - (C) 240 g; photosynthesis
 - (D) 60 g; photosynthesis
 - (E) 38.4 g; photosynthesis
 - (F) 60 g; respiration
- 26.** At night Leh is -13°C and Shimla is 4°C . How much colder is Leh than Shimla, and which place is colder? [\[Integers\]](#)
- (A) 9°C ; Leh
 - (B) -17°C ; Leh
 - (C) 13°C ; Leh
 - (D) 17°C ; Shimla
 - (E) 17°C ; both same
 - (F) 21°C ; Leh
 - (G) 17°C ; Leh
- 27.** From 500 g of hot copper sulphate solution, 120 g of crystals form on cooling. What percent of the solution became crystals, and is crystallisation a physical or a chemical change? [\[Physical & Chemical Changes\]](#)
- (A) 48%; physical
 - (B) 24%; chemical
 - (C) 12%; physical
 - (D) 24%; physical
 - (E) 24 g; physical
 - (F) 20%; physical
- 28.** A circular garden has a radius of 14 m. What is its area? (Take $\pi = \frac{22}{7}$.) [\[Perimeter & Area\]](#)
- (A) 154 sq m
 - (B) 308 sq m
 - (C) 616 sq m
 - (D) 196 sq m
 - (E) 44 sq m
 - (F) 1232 sq m
 - (G) 88 sq m

29. To test a leaf for starch we boil it in alcohol to remove the green colour, then add iodine. A variegated leaf is 60% green, and only the green part makes starch. If the whole leaf is 40 sq cm, what area turns blue-black, and what does blue-black show? [\[Nutrition in Plants\]](#)

- (A) 24 sq cm; no starch
- (B) 24 sq cm; oxygen is present
- (C) 24 sq cm; starch is present
- (D) 6 sq cm; starch is present
- (E) 40 sq cm; starch is present
- (F) 36 sq cm; starch is present

30. A shop's daily results for a week in rupees are +120, -80, +200, -150, +60, -40, +90. What is the net result for the week? [\[Integers\]](#)

- (A) Rs 270 loss
- (B) Rs 740 profit
- (C) Rs 200 profit
- (D) Rs 100 profit
- (E) Rs 470 profit
- (F) Rs 200 loss

31. Milk turns sour because bacteria change it. If 4 of every 5 litres left out turn sour, how many litres out of 35 turn sour, and is souring a physical or a chemical change? [\[Physical & Chemical Changes\]](#)

- (A) 30 L; chemical
- (B) 32 L; chemical
- (C) 28 L; physical
- (D) 28 L; both
- (E) 28 L; chemical
- (F) 7 L; chemical

32. A rectangular sheet is 20 cm by 15 cm. A border 3 cm wide is cut off all around and removed. What is the area of the remaining inner rectangle? [\[Perimeter & Area\]](#)

- (A) 119 sq cm
- (B) 126 sq cm
- (C) 98 sq cm
- (D) 210 sq cm
- (E) 174 sq cm
- (F) 300 sq cm

33. A fertiliser bag is marked nitrogen : phosphorus : potassium = 5 : 3 : 2. If the bag holds 50 kg, how much nitrogen does it contain, and which of these nutrients keeps leaves green? [\[Nutrition in Plants\]](#)

- (A) 15 kg; nitrogen
- (B) 25 kg; potassium
- (C) 25 kg; nitrogen
- (D) 10 kg; nitrogen
- (E) 5 kg; nitrogen
- (F) 25 kg; phosphorus
- (G) 30 kg; nitrogen

- 34.** Multiply (-1) by itself 7 times in a row, then separately multiply (-1) by itself 8 times. What are the two results? [\[Integers\]](#)
- (A) 7 and 8
 - (B) -1 and $+1$
 - (C) $+1$ and $+1$
 - (D) -1 and -1
 - (E) -7 and 8
 - (F) -1 and 0
 - (G) $+1$ and -1
- 35.** You cut a 30 cm paper strip into 6 equal pieces and then burn them. How long is each piece, and how should cutting and burning be classified? [\[Physical & Chemical Changes\]](#)
- (A) 5 cm; both physical
 - (B) 5 cm; cutting physical, burning physical
 - (C) 5 cm; both chemical
 - (D) 24 cm; cutting physical, burning chemical
 - (E) 5 cm; cutting chemical, burning physical
 - (F) 6 cm; cutting physical, burning chemical
 - (G) 5 cm; cutting physical, burning chemical
- 36.** A wall is 6 m long and 3 m high. Painting costs Rs 25 per square metre, plus a 10% extra charge on the painting cost. What is the total cost? [\[Perimeter & Area\]](#)
- (A) Rs 450
 - (B) Rs 405
 - (C) Rs 540
 - (D) Rs 495
 - (E) Rs 472.50
 - (F) Rs 459
 - (G) Rs 500
- 37.** Among 8 living things, 3 are green plants and the rest cannot make their own food. What fraction cannot make their own food, and what are food-making organisms called? [\[Nutrition in Plants\]](#)
- (A) $\frac{5}{8}$; autotrophs
 - (B) $\frac{5}{8}$; saprotrophs
 - (C) $\frac{3}{8}$; autotrophs
 - (D) $\frac{5}{8}$; parasites
 - (E) $\frac{5}{8}$; heterotrophs
 - (F) $\frac{3}{5}$; autotrophs
 - (G) $\frac{1}{2}$; autotrophs
- 38.** Take -144 , divide it by -12 , then multiply by -2 , then add -6 . What is the final result? [\[Integers\]](#)
- (A) -24
 - (B) -18
 - (C) -30
 - (D) 6
 - (E) -36
 - (F) 30
 - (G) 18

- 39.** Water expands a little when heated, so 1000 mL becomes 1004 mL when warmed. What is the percent increase in volume, and is this a physical or a chemical change? [\[Physical & Chemical Changes\]](#)
- (A) 40%; physical
 - (B) 4%; physical
 - (C) 0.4 mL; physical
 - (D) 0.04%; physical
 - (E) 1.004%; physical
 - (F) 0.4%; physical
 - (G) 0.4%; chemical
- 40.** An L-shaped room is made from an 8 m by 5 m rectangle with a 3 m by 2 m corner removed. What is the area of the room? [\[Perimeter & Area\]](#)
- (A) 38 sq m
 - (B) 32 sq m
 - (C) 6 sq m
 - (D) 34 sq m
 - (E) 40 sq m
 - (F) 26 sq m
- 41.** One leaf makes 0.5 g of glucose a day, but one-quarter of a plant's 24 leaves sit in shade and make only half as much. How much glucose does the whole plant make in a day, and what pigment captures the light? [\[Nutrition in Plants\]](#)
- (A) 12 g; chlorophyll
 - (B) 10.5 g; chlorophyll
 - (C) 7.5 g; chlorophyll
 - (D) 10.5 g; cytoplasm
 - (E) 9 g; chlorophyll
 - (F) 10.5 g; chloroplast
- 42.** A lift starts at floor +9. It goes down 4 floors, then down 8 more into the basement, then up 5. Floors below the ground are negative. Where does it stop? [\[Integers\]](#)
- (A) floor +2
 - (B) floor +12
 - (C) floor +6
 - (D) floor +10
 - (E) floor -2
 - (F) floor -3
 - (G) floor 0
- 43.** Mixing baking soda with vinegar fizzes and gives off a gas. If 2 g of soda gives 1 g of gas, how much gas comes from 9 g of soda, and what kind of change is the fizzing? [\[Physical & Chemical Changes\]](#)
- (A) 4.5 g; physical
 - (B) 18 g; chemical
 - (C) 4.5 g; both
 - (D) 9 g; chemical
 - (E) 2.25 g; chemical
 - (F) 4.5 g; chemical

44. A rectangular plot 30 m by 20 m is fenced, but a 4 m wide gate needs no fence. Fencing costs Rs 50 per metre. What is the cost of the fence? [\[Perimeter & Area\]](#)

- (A) Rs 5000
- (B) Rs 5200
- (C) Rs 4900
- (D) Rs 4800
- (E) Rs 3000
- (F) Rs 2400

45. In a lichen, a fungus and an alga live together: the alga makes food while the fungus gives water and shelter. If 35 of 50 lichens on a wall survive a dry month, what percent survive, and what is this living-together called? [\[Nutrition in Plants\]](#)

- (A) 70%; symbiosis
- (B) 70%; parasitism
- (C) 70%; competition
- (D) 30%; symbiosis
- (E) 65%; symbiosis
- (F) 85%; symbiosis
- (G) 70%; saprotrophism

46. The midday temperatures for five days were -2, 5, -4, 6 and 0 (deg C). What is their sum, and is that total positive or negative? [\[Integers\]](#)

- (A) -1; negative
- (B) 5; negative
- (C) 5; zero
- (D) 5; positive
- (E) -5; positive
- (F) 1; positive
- (G) 17; positive

47. Which list contains only changes that can be reversed: (i) melting ice, freezing water, dissolving salt; or (ii) burning wood, cooking an egg, rusting iron? [\[Physical & Chemical Changes\]](#)

- (A) list (ii)
- (B) only melting ice
- (C) only freezing water
- (D) neither list
- (E) both lists
- (F) list (i)

48. A triangle has an area of 84 sq cm and a base of 24 cm. What is its height? [\[Perimeter & Area\]](#)

- (A) 8 cm
- (B) 60 cm
- (C) 3 cm
- (D) 14 cm
- (E) 7 cm
- (F) 10 cm
- (G) 3.5 cm

- 49.** Air is about 78% nitrogen, but plants cannot use it straight from the air. From a 500-litre sample of air, how many litres are nitrogen, and what changes air-nitrogen into a form plants can use? [\[Nutrition in Plants\]](#)
- (A) 110 L; nitrogen-fixing bacteria
 - (B) 420 L; nitrogen-fixing bacteria
 - (C) 78 L; nitrogen-fixing bacteria
 - (D) 390 L; oxygen
 - (E) 390 L; nitrogen-fixing bacteria
 - (F) 390 L; chlorophyll
 - (G) 39 L; nitrogen-fixing bacteria
- 50.** A number multiplied by -6 gives 42. What is the number, and is it positive or negative? [\[Integers\]](#)
- (A) -7; positive
 - (B) -8; negative
 - (C) -7; negative
 - (D) 36; negative
 - (E) -6; negative
 - (F) 7; negative
 - (G) 7; positive
- 51.** Iron nails are kept in three test tubes: A in dry air, B in boiled water with no air, and C in air plus water. In how many tubes does rust form, and which tube? [\[Physical & Chemical Changes\]](#)
- (A) 1; tube C
 - (B) 3; all of them
 - (C) 2; tubes B and C
 - (D) 1; tube A
 - (E) 2; tubes A and C
 - (F) 1; tube B
- 52.** A big square has a side of 10 cm. A smaller square of side 6 cm sits inside it. What area lies outside the small square but inside the big one? [\[Perimeter & Area\]](#)
- (A) 40 sq cm
 - (B) 49 sq cm
 - (C) 64 sq cm
 - (D) 16 sq cm
 - (E) 136 sq cm
 - (F) 60 sq cm
 - (G) 36 sq cm
- 53.** Five glucose-making cells each produce 8 units of glucose, and three-quarters of all that glucose is turned into stored food. How many units are stored, and in what form is the food stored? [\[Nutrition in Plants\]](#)
- (A) 30 units; glucose
 - (B) 32 units; starch
 - (C) 40 units; starch
 - (D) 20 units; starch
 - (E) 30 units; protein
 - (F) 30 units; starch

54. Compare $(-15 + 20)$ with $(8 - 14)$. Which is greater, and by how much? [\[Integers\]](#)

- (A) the second; greater by 11
- (B) the first; greater by 11
- (C) the first; greater by 6
- (D) the first; greater by 14
- (E) the second; greater by 1
- (F) they are equal

55. When 16 g of methane burns, it joins with 64 g of oxygen. By conservation of mass, what is the total mass of the products, and is burning a physical or a chemical change? [\[Physical & Chemical Changes\]](#)

- (A) 80 g; physical
- (B) 1024 g; chemical
- (C) 80 g; chemical
- (D) 80 g; both
- (E) 48 g; chemical
- (F) 64 g; chemical

56. A rectangle's length is 3 times its width, and its perimeter is 64 cm. What is its area? [\[Perimeter & Area\]](#)

- (A) 192 sq cm
- (B) 256 sq cm
- (C) 48 sq cm
- (D) 144 sq cm
- (E) 64 sq cm
- (F) 96 sq cm

57. A water plant gives off 15 oxygen bubbles a minute in bright light, but only one-third as many in dim light. How many bubbles does it give off in 4 minutes of dim light, and what gas are the bubbles? [\[Nutrition in Plants\]](#)

- (A) 20 bubbles; carbon dioxide
- (B) 60 bubbles; oxygen
- (C) 5 bubbles; oxygen
- (D) 20 bubbles; oxygen
- (E) 45 bubbles; oxygen
- (F) 12 bubbles; oxygen
- (G) 20 bubbles; hydrogen

58. Use a short method to evaluate $(-18) \times 7 + (-18) \times 3$. [\[Integers\]](#)

- (A) -90
- (B) -126
- (C) 180
- (D) -54
- (E) -180
- (F) -100

59. Sea water holds 35 g of salt per litre. If 8 litres are evaporated, how much salt is left, and is getting salt this way a physical or a chemical change? [\[Physical & Chemical Changes\]](#)

- (A) 280 g; chemical
- (B) 280 g; both
- (C) 245 g; physical
- (D) 280 g; physical
- (E) 43 g; physical
- (F) 4.375 g; physical
- (G) 2800 g; physical

60. A bicycle wheel covers 132 cm in one full turn. What is the wheel's diameter? (Take $\pi = 22/7$.) [\[Perimeter & Area\]](#)

- (A) 14 cm
- (B) 21 cm
- (C) 6 cm
- (D) 12 cm
- (E) 42 cm
- (F) 66 cm
- (G) 84 cm

61. A neem leaf is roughly a triangle with base 2 cm and height 5 cm. A twig carries 20 such leaves. What is the total leaf area on the twig, and what is the food-making in these leaves called? [\[Nutrition in Plants\]](#)

- (A) 50 sq cm; photosynthesis
- (B) 10 sq cm; photosynthesis
- (C) 100 sq cm; transpiration
- (D) 100 sq cm; respiration
- (E) 100 sq cm; photosynthesis
- (F) 140 sq cm; photosynthesis
- (G) 200 sq cm; photosynthesis

62. How many integers lie strictly between -5 and 4, not counting -5 and 4 themselves? [\[Integers\]](#)

- (A) 8
- (B) 9
- (C) 10
- (D) 11
- (E) 7
- (F) 6
- (G) 2

63. Curd sets when a starter is added to milk, using 1 spoon for every 250 mL. For 4.5 litres of milk, how many spoons are needed, and is curd setting a physical or a chemical change? [\[Physical & Chemical Changes\]](#)

- (A) 36 spoons; chemical
- (B) 18 spoons; both
- (C) 9 spoons; chemical
- (D) 18 spoons; chemical
- (E) 4.5 spoons; chemical
- (F) 45 spoons; chemical

64. A rectangle 16 cm by 9 cm is cut along a diagonal into two equal triangles. What is the area of one triangle? [\[Perimeter & Area\]](#)

- (A) 48 sq cm
- (B) 36 sq cm
- (C) 144 sq cm
- (D) 50 sq cm
- (E) 72 sq cm
- (F) 25 sq cm
- (G) 288 sq cm

65. A farmer adds 2.5 kg of compost to each of 16 plant beds. Compost returns nutrients to the soil. How much compost is used in all, and is compost a natural manure or a chemical fertiliser? [\[Nutrition in Plants\]](#)

- (A) 6.4 kg; natural manure
- (B) 40 g; natural manure
- (C) 400 kg; natural manure
- (D) 20 kg; natural manure
- (E) 40 kg; natural manure
- (F) 40 kg; chemical fertiliser
- (G) 18.5 kg; natural manure

66. Riya owes Rs 250, shown as -250. She then earns Rs 90 three times and spends Rs 60. What is her balance now? [\[Integers\]](#)

- (A) -340 (owes Rs 340)
- (B) -40 (owes Rs 40)
- (C) -20 (owes Rs 20)
- (D) +40 (has Rs 40)
- (E) +20 (has Rs 20)
- (F) -160 (owes Rs 160)

67. A 12 cm magnesium ribbon burns at 3 cm per second, leaving white ash. How long does it take to burn fully, and is this a physical or a chemical change? [\[Physical & Chemical Changes\]](#)

- (A) 9 seconds; chemical
- (B) 4 seconds; chemical
- (C) 4 seconds; physical
- (D) 0.25 seconds; chemical
- (E) 6 seconds; chemical
- (F) 36 seconds; chemical

68. A courtyard is 12 m by 9 m and is paved with square slabs of side 60 cm. How many slabs are needed? [\[Perimeter & Area\]](#)

- (A) 30 slabs
- (B) 108 slabs
- (C) 360 slabs
- (D) 250 slabs
- (E) 3000 slabs
- (F) 300 slabs
- (G) 180 slabs

69. Of 12 organisms studied, 5 are green plants, 4 are fungi feeding on dead matter, and 3 are parasites on living hosts. What fraction are heterotrophs (cannot make their own food), and what are the green plants called? [\[Nutrition in Plants\]](#)

- (A) $\frac{7}{12}$; autotrophs
- (B) $\frac{7}{12}$; heterotrophs
- (C) $\frac{5}{7}$; autotrophs
- (D) $\frac{1}{2}$; autotrophs
- (E) $\frac{7}{12}$; parasites
- (F) $\frac{5}{12}$; autotrophs

70. Evaluate the expression $5 - [3 - (-2 + 6)]$. [\[Integers\]](#)

- (A) 8
- (B) 0
- (C) 4
- (D) 2
- (E) 6
- (F) -1

71. A 24 cm wire is bent into a square. What is the side length, and is bending a physical or a chemical change? [\[Physical & Chemical Changes\]](#)

- (A) 8 cm; physical
- (B) 6 cm; both
- (C) 4 cm; physical
- (D) 12 cm; physical
- (E) 6 cm; chemical
- (F) 6 cm; physical
- (G) 24 cm; physical

72. A square and a rectangle have the same perimeter of 48 cm, and the rectangle is 16 cm long. Which shape has the bigger area, and by how much? [\[Perimeter & Area\]](#)

- (A) the square, by 16 sq cm
- (B) the square, by 32 sq cm
- (C) the rectangle, by 8 sq cm
- (D) the square, by 8 sq cm
- (E) the rectangle, by 32 sq cm
- (F) the rectangle, by 16 sq cm

73. A leaf makes 36 mg of glucose and sends five-sixths of it to the rest of the plant, keeping the rest. How many milligrams does the leaf keep, and through which tissue is the food carried away? [\[Nutrition in Plants\]](#)

- (A) 6 mg; phloem
- (B) 30 mg; phloem
- (C) 6 mg; stomata
- (D) 5 mg; phloem
- (E) 6 mg; xylem
- (F) 36 mg; phloem

74. Find the sign and the value of the product $(-2) \times 3 \times (-4) \times (-1) \times 5$. [\[Integers\]](#)

- (A) negative; -100
- (B) positive; -120
- (C) negative; -60
- (D) positive; 120
- (E) negative; -120
- (F) negative; 120
- (G) negative; -240

75. Heating blue copper sulphate crystals drives off water and turns them white; adding water turns them blue again. If 50 g loses 18 g of water, what mass stays white, and can this change be reversed? [\[Physical & Chemical Changes\]](#)

- (A) 32 g; yes, reversible
- (B) 18 g; yes, reversible
- (C) 32 g; only by burning
- (D) 32 g; no, not reversible
- (E) 36 g; yes, reversible
- (F) 68 g; yes, reversible

76. If the side of a square is doubled, how many times bigger does its area become? Check using a side of 5 cm. [\[Perimeter & Area\]](#)

- (A) 16 times
- (B) 6 times
- (C) 8 times
- (D) 4 times
- (E) 3 times
- (F) 2 times

77. Before a starch test a plant is kept in the dark for 48 hours. It makes 4 mg of starch an hour in light but uses 1 mg an hour in the dark. If it starts with 80 mg of starch, how much is left after the 48 dark hours, and why is the dark step done? [\[Nutrition in Plants\]](#)

- (A) 0 mg; to use up starch already there
- (B) 32 mg; to give the leaf light
- (C) 32 mg; to add starch to the leaf
- (D) 32 mg; to use up starch already there
- (E) 52 mg; to use up starch already there
- (F) 48 mg; to use up starch already there
- (G) 16 mg; to use up starch already there

78. A freezer cools at a steady 4 deg C drop each hour, starting from 15 deg C. After how many hours does it reach -13 deg C? [\[Integers\]](#)

- (A) 3 hours
- (B) 2 hours
- (C) 7 hours
- (D) 4 hours
- (E) 28 hours
- (F) 8 hours
- (G) 12 hours

79. Digesting food is a chemical change that releases energy. If one biscuit gives 90 units of energy and you eat 2.5 biscuits, how much energy is released, and is digestion a physical or a chemical change?

[Physical & Chemical Changes]

- (A) 225 units; chemical
- (B) 225 units; physical
- (C) 225 units; both
- (D) 36 units; chemical
- (E) 250 units; chemical
- (F) 92.5 units; chemical

80. A field is 12.5 m long and 8 m wide. What are its area and its perimeter? [Perimeter & Area]

- (A) 100 sq m; 20.5 m
- (B) 20.5 sq m; 41 m
- (C) 112.5 sq m; 41 m
- (D) 100 sq m; 82 m
- (E) 96 sq m; 41 m
- (F) 100 sq m; 40 m
- (G) 100 sq m; 41 m

81. Plant A makes 3 mg of glucose an hour and plant B makes 2 mg an hour. After how many hours will plant A have made 8 mg more than plant B, and what raw materials does a plant need to make glucose?

[Nutrition in Plants]

- (A) 4 hours; carbon dioxide and water
- (B) 8 hours; carbon dioxide and water
- (C) 2 hours; carbon dioxide and water
- (D) 16 hours; carbon dioxide and water
- (E) 8 hours; carbon dioxide and oxygen
- (F) 8 hours; nitrogen and water
- (G) 8 hours; oxygen and water

82. By how much is the product $(-9) \times 8$ less than the product $(-6) \times (-7)$? [Integers]

- (A) 102
- (B) 114
- (C) 30
- (D) 42
- (E) -30
- (F) 72
- (G) -114

83. A green fruit ripens to yellow, which is a chemical change. If a basket has 60 fruits and 3 ripen each day, after how many days are all ripe, and is ripening a physical or a chemical change? [Physical & Chemical

Changes]

- (A) 30 days; chemical
- (B) 20 days; chemical
- (C) 20 days; physical
- (D) 20 days; both
- (E) 57 days; chemical
- (F) 180 days; chemical

- 84.** A rectangle is 20 cm by 12 cm. Inside it a triangle has the full 20 cm base and a height of 12 cm. What area of the rectangle lies outside the triangle? [\[Perimeter & Area\]](#)
- (A) 32 sq cm
 - (B) 240 sq cm
 - (C) 100 sq cm
 - (D) 80 sq cm
 - (E) 120 sq cm
 - (F) 60 sq cm
- 85.** Without manure a field yields 200 kg of grain. With manure the yield rises by 35%. What is the new yield, and manure improves the soil chiefly by adding what? [\[Nutrition in Plants\]](#)
- (A) 130 kg; nutrients
 - (B) 300 kg; nutrients
 - (C) 235 kg; nutrients
 - (D) 270 kg; nutrients
 - (E) 70 kg; nutrients
 - (F) 270 kg; water only
 - (G) 207 kg; nutrients
- 86.** A debt of Rs -180 is shared equally among 6 friends. What does each friend owe, and is that a positive or negative amount? [\[Integers\]](#)
- (A) Rs 30 each (positive)
 - (B) Rs -174 each (negative)
 - (C) Rs -36 each (negative)
 - (D) Rs -3 each (negative)
 - (E) Rs -30 each (negative)
 - (F) Rs -1080 each (negative)
- 87.** Water expands about 10% when it freezes. A pipe holds 500 mL of water. What volume does the ice take up, and is freezing a physical or a chemical change? [\[Physical & Chemical Changes\]](#)
- (A) 550 mL; physical
 - (B) 510 mL; physical
 - (C) 550 mL; both
 - (D) 550 mL; chemical
 - (E) 600 mL; physical
 - (F) 450 mL; physical
 - (G) 50 mL; physical
- 88.** A circular park has a radius of 21 m. Fencing costs Rs 15 per metre. What is the cost to fence it once all the way around? (Take $\pi = 22/7$.) [\[Perimeter & Area\]](#)
- (A) Rs 1980
 - (B) Rs 2100
 - (C) Rs 990
 - (D) Rs 1386
 - (E) Rs 2970
 - (F) Rs 1320
 - (G) Rs 660

89. A Venus flytrap snaps shut only after a trigger hair is touched twice within 20 seconds. An insect brushes a hair once every 8 seconds. How many touches happen within 20 seconds, and does the trap close? [\[Nutrition in Plants\]](#)

- (A) 3 touches; yes, it closes
- (B) 2 touches; no, it needs three
- (C) 1 touch; no, it stays open
- (D) 2 touches; no, it stays open
- (E) 2 touches; yes, it closes
- (F) 4 touches; yes, it closes

90. Start at 0 and follow this pattern: subtract 7, multiply by 3, add 11, then subtract 5. What number do you end on? [\[Integers\]](#)

- (A) 15
- (B) -21
- (C) -10
- (D) -9
- (E) -31
- (F) -25
- (G) -15

91. Which signs together best show that a chemical change has happened: (i) a change of shape only; or (ii) a new colour, a gas given off, and heat or light produced? [\[Physical & Chemical Changes\]](#)

- (A) only getting colder
- (B) a change of shape only (i)
- (C) both equally
- (D) neither one
- (E) only a change of state
- (F) the new colour, gas, and heat or light (ii)
- (G) only a change of size

92. Carpet costs Rs 240 per square metre. A room is 5 m by 4.5 m, and there is a 5% discount on the total. What is the final cost? [\[Perimeter & Area\]](#)

- (A) Rs 4860
- (B) Rs 5130
- (C) Rs 5270
- (D) Rs 5160
- (E) Rs 5400
- (F) Rs 270

93. A plant's rate of photosynthesis is proportional to the light it gets. At 100 units of light it makes 8 mg of glucose an hour. How much will it make at 250 units of light, and what gas does it give off? [\[Nutrition in Plants\]](#)

- (A) 20 mg; oxygen
- (B) 25 mg; oxygen
- (C) 12.5 mg; oxygen
- (D) 20 mg; carbon dioxide
- (E) 3.2 mg; oxygen
- (F) 16 mg; oxygen

94. Arrange -3, 0, -7, 5 and -1 from lowest to highest. Which value is in the middle? [\[Integers\]](#)

- (A) -1
- (B) 1
- (C) 5
- (D) -3
- (E) -2
- (F) -7
- (G) 0

95. An iron sheet loses 6 g to rust in 4 months. At the same rate, how much rusts in 10 months, and which two things together cause rust? [\[Physical & Chemical Changes\]](#)

- (A) 2.4 g; air and moisture
- (B) 15 g; air and oil
- (C) 15 g; heat and light
- (D) 24 g; air and moisture
- (E) 12 g; air and moisture
- (F) 15 g; air and moisture

96. A square tile has an area of 169 sq cm. What is its perimeter? [\[Perimeter & Area\]](#)

- (A) 52 cm
- (B) 13 cm
- (C) 169 cm
- (D) 48 cm
- (E) 42.25 cm
- (F) 26 cm
- (G) 676 cm

97. In a sealed jar a plant releases 30 mL of oxygen an hour while a mouse uses 24 mL of oxygen an hour. What net volume of oxygen is added to the jar in 5 hours, and which process of the plant supplies it? [\[Nutrition](#)

[in Plants\]](#)

- (A) 30 mL; photosynthesis
- (B) 150 mL; photosynthesis
- (C) 6 mL; photosynthesis
- (D) 30 mL; respiration
- (E) 120 mL; photosynthesis
- (F) 270 mL; photosynthesis

98. In a board game, moving right is positive. A token moves +6, -9, +4, -2 and +3. Where does it end relative to the start, and in which direction? [\[Integers\]](#)

- (A) -2; to the right
- (B) -2; to the left
- (C) +8; to the right
- (D) +2; to the left
- (E) +2; no movement
- (F) +2; to the right

99. A mixture has 40 g of iron filings and 60 g of sand. A magnet removes all the iron. What percent of the mixture did the magnet remove, and is separating it a physical or a chemical change? [\[Physical & Chemical Changes\]](#)

- (A) 20%; physical
- (B) 24%; physical
- (C) 40%; chemical
- (D) 40 g; physical
- (E) 100%; physical
- (F) 40%; physical

100. Room A is 6 m by 5 m and room B is 4 m by 3 m. Flooring costs Rs 80 per square metre. What is the total cost to floor both rooms? [\[Perimeter & Area\]](#)

- (A) Rs 2400
- (B) Rs 1680
- (C) Rs 3200
- (D) Rs 960
- (E) Rs 3360
- (F) Rs 3440
- (G) Rs 3300